Multi-Antenna Wi-Fi Activity Recognition with Doppler Spectrograms

Barış Kalfa

***Abstract*—Wi-Fi channel measurements can support privacy-preserving human activity recognition without cameras or wearable sensors. This report reproduces the SHARP processing and learning framework using precomputed Doppler traces. A lightweight multi-scale convolutional neural network classifies five activities from one antenna at a time; four independent predictions are then combined through the paper's decision-fusion rule. The PyTorch implementation matches the published architecture's 128,535 parameters and uses a chronological 60/20/20 split of the S1 recordings for training, validation, and reserved test data. The frozen final model achieves 100% fused accuracy on the reserved S1 split and 94.35–99.87% across the external S2–S7 settings. Under a stricter S1a/S1b-to-S1c split, neither time-shift augmentation (83.21% fused accuracy) nor a training-only supervised-contrastive projection head (82.17%) improves on the matched supervised baseline (84.23%). This controlled result highlights the remaining subject/session shift.**

***Index Terms*—Wi-Fi sensing, human activity recognition, Doppler spectrograms, convolutional neural networks, decision fusion.**

I. Introduction

Human activity recognition (HAR) is useful for assisted living, security, and smart spaces, but camera-based systems raise privacy concerns and wearable devices require user cooperation. Wi-Fi sensing offers a complementary approach: a person's motion changes the wireless channel, and these changes can be interpreted without recording images. In particular, Doppler spectrograms expose motion-dependent frequency shifts that can be processed as two-dimensional signals.

This project studies HAR from the precomputed Doppler traces released with the SHARP framework [1]. We focus on five classes: empty room, sitting, walking, running, and jumping. The work has two goals. First, we reproduce the compact single-antenna architecture and multi-antenna decision fusion described in SHARP. Second, we evaluate the frozen model on the reserved portion of S1 and on the external S2–S7 measurement settings. This separates model selection from the final comparison.

Our implementation retains the paper's model structure and fusion rule while making the experimental protocol explicit. The main contributions are:

- a PyTorch implementation of the 128,535-parameter SHARP classifier;
- chronological splits, Doppler-window construction, and optimization settings aligned with the published reference code;
- a reproducible evaluation that reports both per-antenna and four-antenna fused performance on S1–S7.

Section II summarizes the dataset and SHARP method. Section III describes the signals, network, and learning protocol. Section IV presents the experimental results, followed by conclusions in Section V.

II. Background and Related Work

Wi-Fi sensing estimates changes in the channel frequency response (CFR) caused by moving people. The dataset paper by Meneghello *et al.* [2] provides 80 MHz IEEE 802.11ac measurements with diversity in people, rooms, days, monitor positions, and direct-path conditions. Each monitor has four receiving antennas. The dataset is designed to test whether a learning system can generalize beyond the measurement condition used for training.

SHARP [1] turns sanitized CFR measurements into Doppler traces and frames HAR as a five-class classification problem. It applies the same compact CNN to each antenna separately and combines their decisions only after classification. This avoids assigning meaning to the order of antennas, which the authors found unsuitable for an environment-invariant input-level fusion model. SHARP reports near-perfect performance in several settings, with the harder blocked-path and unseen-environment cases exposing most of the remaining walking–running ambiguity.

The present work is a reproduction baseline rather than a new architecture. It uses the supplied precomputed Doppler traces, so raw-CFR normalization, phase sanitization, and Doppler extraction are inherited from the released data. The focus is therefore on the learnable classification and decision-fusion stages.

III. Processing and Learning Framework

A. Doppler traces and data partition

Each input is a 340×100 matrix: 340 consecutive Doppler vectors (about two seconds) by 100 velocity bins. A trace has one such matrix per candidate time window and per antenna. The stored values are normalized linear powers. For every antenna recording, we subtract the temporal mean of each Doppler bin before extracting windows. This matches the offline preprocessing in the reference implementation; consequently, the transform uses the statistics of the complete recording and is not causal.

The S1a, S1b, and S1c recordings are each divided chronologically into 60% training, 20% validation, and the remaining 20% reserved S1 test data. A 31-row gap separates adjacent

portions. Windows contain 340 rows and advance by 30 rows, following the authors' code. The other sessions, S2–S7, are never used to select model parameters. They are evaluated only after the final checkpoint is frozen. Classes $\{E, L, W, R, J\}$ denote empty room, sitting, walking, running, and jumping.

B. Single-antenna classifier

The classifier takes a tensor of shape $1 \times 340 \times 100$. Its first stage is a simplified Inception-style reduction block with three parallel branches: max pooling with 2×2 kernel and stride two; a 2×2 convolution producing five feature maps; and a sequence of 1×1, 2×2, and 4×4 convolutions producing 3, 6, and 9 maps, respectively. The branch outputs have common spatial resolution and are concatenated into 15 maps. A 1×1 convolution reduces these to three maps. ReLU follows each convolution.

The resulting $3 \times 170 \times 50$ tensor is flattened, regularized with dropout probability 0.2, and mapped to five unnormalized class scores.

The network contains 128,535 trainable parameters, exactly the number reported for SHARP [1]. Training minimizes the sparse cross-entropy loss

$$\mathcal{L} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(z_{i,y_i})}{\sum_{c=1}^{5} \exp(z_{i,c})}, \tag{1}$$

where $z_{i,c}$ is the logit for class c and y_i is the true class. We use Adam with learning rate 10^{-4}, batches of 32, at most 25 epochs, and early stopping on validation loss with patience three. The final epoch checkpoint is retained to follow the behavior of the published training script.

C. Four-antenna decision fusion

For an aligned window, antenna a produces an activity vector $\boldsymbol{z}_a \in \mathbb{R}^5$ and label $\hat{y}_a = \arg\max_c z_{a,c}$. When at least three of the four labels agree, that class is selected. Otherwise, the fused label is

$$\hat{y} = \arg\max_{c} \sum_{a=1}^{4} z_{a,c}. \tag{2}$$

Thus, softmax is used only implicitly by the cross-entropy loss during training; the fallback fusion sums raw output scores, as specified by SHARP. This rule preserves independent per-antenna inference while exploiting agreement among antennas.

D. Held-out-session contrastive experiment

To test whether representation learning improves robustness, we kept the deployed CNN unchanged and attached a temporary projection head ($25500 \to 128 \to 64$) only during pretraining. The contrastive loss treats the four antenna views of the same physical window as positives and also treats same-activity windows from S1a and S1b as positives. Two augmented views use small zero-filled time shifts, gain/noise perturbations, and limited time/frequency masking. S1c is excluded entirely from this procedure and used only for validation.

TABLE I
FROZEN-CHECKPOINT EVALUATION. S1 IS THE RESERVED CHRONOLOGICAL SPLIT; S2–S7 ARE EXTERNAL MEASUREMENT SETTINGS.

Set	Windows	Single acc.	Fused acc.	Fused F1
S1	1,636	95.94%	100.00%	1.000
S2	7,924	93.55%	99.87%	0.998
S3	3,095	84.71%	99.77%	0.998
S4	6,111	84.30%	96.25%	0.962
S5	3,062	78.78%	94.35%	0.943
S6	6,149	88.42%	99.69%	0.997
S7	2,872	84.14%	95.51%	0.958

a) Training-only projection head.: The projection head is discarded before standard classifier fine-tuning. It is therefore absent from Figure 1, validation, external testing, and four-antenna decision fusion: deployment uses only the original CNN, dropout, and five-class classifier.

b) Time-shift control.: To isolate the value of the contrastive objective from its most direct invariance, we trained the same supervised CNN with only the zero-filled ± 15-row time-shift augmentation. It uses the same S1a/S1b training data, S1c validation recording, seed, optimizer, stride, and early-stopping rule as the baseline and contrastive runs.

IV. EXPERIMENTAL RESULTS

The final model stopped after 23 epochs. On the chronological validation portions of S1, individual antenna windows achieved 95.92% accuracy and 0.959 macro F1. Applying four-antenna decision fusion increased validation accuracy to 100%. This validation setting contains overlapping windows from the same recordings as training, so it primarily verifies agreement with the published protocol rather than robustness to a new environment.

The frozen checkpoint was then evaluated once on the reserved S1 portions and on S2–S7. Table I compares individual-antenna classification with decision fusion. "Single" weights every antenna-window prediction equally; "fused" produces one label per aligned four-antenna window. The final column reports macro F1, the unweighted mean of the five per-class F1 scores.

Fusion improves every tested set and is particularly important in the difficult S4 and S5 conditions, where individual antenna performance is lower. Figure 2 visualizes that gain across all seven evaluation settings. Figure 3 shows the fused confusion matrices normalized by true class. S5's principal weakness is running, with 72.29% recall. In S7, the unseen-environment condition, walking has 80.91% recall while jumping reaches 100%. These patterns agree with the paper's observation that walking and running are the most easily confused activities.

For a direct comparison with Table III of SHARP, the published mean class accuracies are 99.59%, 99.78%, 99.33%, 97.55%, 96.59%, 99.79%, and 95.99% for S1–S7, respectively [1]. Our corresponding mean recalls are 100.00%, 99.79%, 99.78%, 96.14%, 94.39%, 99.69%, and 95.80%. Five

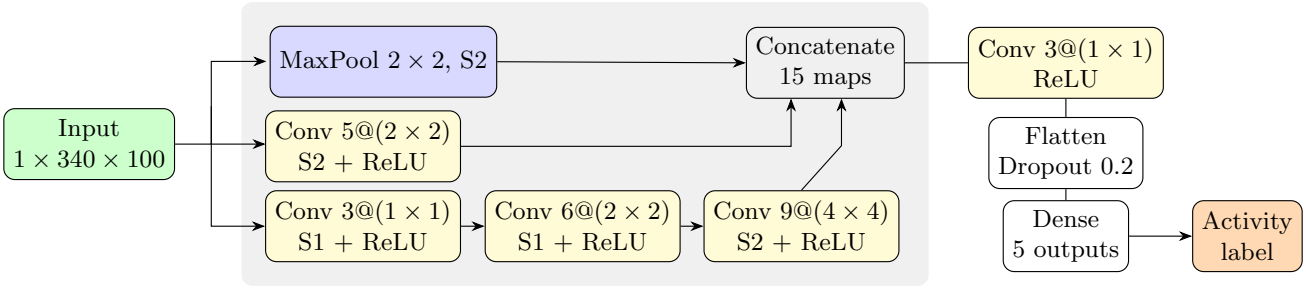


Fig. 1. Implemented SHARP classifier. Three branches reduce the input in parallel, concatenate into 15 maps, then use a 1×1 fusion convolution, flattening, dropout, and a five-output dense classifier.

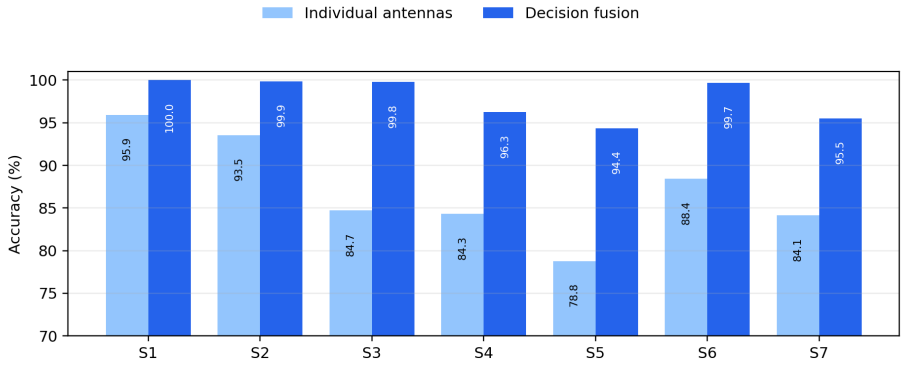


Fig. 2. Frozen-checkpoint accuracy by measurement setting. Four-antenna decision fusion improves every setting over individual-antenna classification.

settings are within 0.5 percentage points; S4 and S5 are lower by 1.41 and 2.20 points. Exact equality is not expected because the original code uses TensorFlow whereas this work uses PyTorch, which changes random-number streams, shuffling, dropout masks, and numerical kernels.

The 30-row stride creates strongly overlapping windows, so neighboring samples are correlated. The reported numbers are therefore useful for reproduction and comparison with SHARP, but they should not be interpreted as independent trial counts. Further development should use held-out recordings for model selection, repeated seeds to quantify variability, and a separate untouched test set for final claims.

A. *Held-out-session comparison*

The chronological protocol can be optimistic because every S1 recording contributes to both training and validation. We therefore performed a stricter comparison: train on all S1a and S1b windows, reserve all S1c windows for validation, and retain a 30-row stride in both sets. Table II compares the supervised baseline, a time-shift-only control, and the domain-aware contrastive variant under this identical split. Both results use each method's best macro-F1 checkpoint and were then fused with the same four-antenna rule.

Decision fusion improves all three methods, but neither added invariance improved cross-session generalization in this run. The time-shift control retained 83.21% fused accuracy, while contrastive pretraining reached 82.17%, both below the unaugmented baseline's 84.23%. This controlled ordering indicates that the contrastive objective, its positive-pair definition, and the associated perturbations did not add useful robustness beyond the original classifier. The contrastive model particularly confused jumping with running: fused jumping recall was 1.83%, compared with 12.61% for time shifts and 14.44% for the supervised baseline. Figure 4 identifies this main failure mode.

TABLE II
ENTIRE-SESSION-HELD-OUT S1C VALIDATION. HIGHER IS BETTER.

Method	Single acc.	Single F1	Fused acc.	Fused F1
Supervised baseline	72.71%	0.729	**84.23%**	**0.793**
Time-shift augmentation	71.92%	0.722	83.21%	0.781
Domain-aware contrastive	71.22%	0.712	82.17%	0.739

The error is systematic rather than an absence of jumping examples: S1c contains 547 aligned jumping windows. Of these, the baseline predicts 466 as running, the time-shift control predicts 478 as running, and the contrastive model predicts 447 as running (with 90 assigned to walking). A likely explanation is that jumping and running both produce short high-velocity Doppler-energy bursts within the same two-second window. Changes in the subject's execution, location, and antenna geometry can alter the temporal detail that separates the two activities, leaving their coarse Doppler patterns

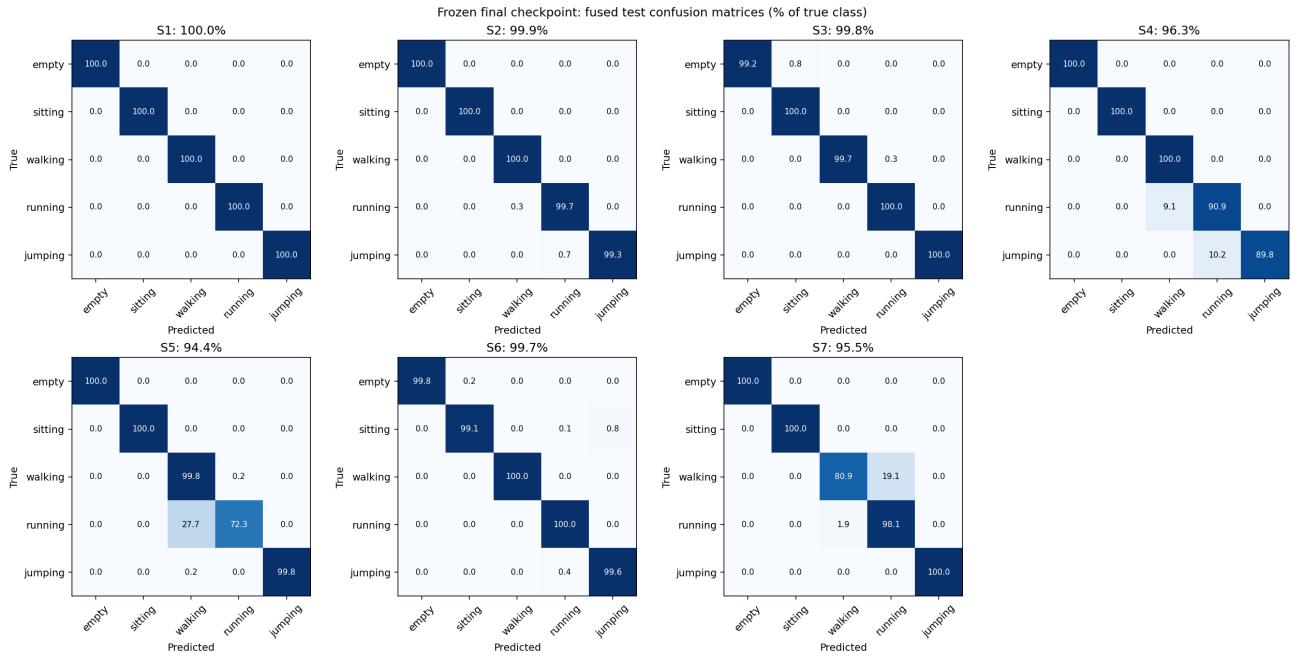


Fig. 3. Fused confusion matrices for the frozen checkpoint. Each matrix is normalized by its true class; values are percentages.

similar. The confusion matrices establish the error pattern; this signal-level mechanism is an interpretation to test with subject- and environment-controlled recordings. Since S1c was used for development selection, the already reported S2–S7 benchmark remains the frozen baseline evaluation rather than evidence for the new methods.

V. Conclusions

This report reproduced a compact multi-antenna Wi-Fi HAR baseline using precomputed Doppler spectrograms. The CNN and decision-fusion rule match SHARP, and the resulting frozen evaluation closely follows the published S1–S7 performance. Fusion consistently improves over a single antenna, while the remaining errors concentrate in walking–running discrimination under difficult propagation conditions.

The main lesson is that experimental protocol matters as much as the architecture: chronological S1 partitions reproduce the reference task, whereas holding out S1c entirely is substantially harder. In the matched S1a/S1b-to-S1c experiment, the unaugmented supervised baseline outperformed both the time-shift control and the implemented domain-aware contrastive method after fusion (84.23%, 83.21%, and 82.17%, respectively). This controlled ablation does not support claiming an improvement from either approach, but it exposes jumping–running transfer as a concrete next target. The current preprocessing also uses complete-recording mean centering, which is appropriate for the offline reference but not directly for streaming deployment. Further work should use repeated held-out sessions, causal normalization, and augmentation tuned without using external test sessions.

References

[1] F. Meneghello, D. Garlisi, N. D. Fabbro, I. Tinnirello, and M. Rossi, "SHARP: Environment and person independent activity recognition with commodity IEEE 802.11 access points," *IEEE Transactions on Mobile Computing*, vol. 22, no. 10, pp. 6160–6175, 2023.

[2] F. Meneghello, N. D. Fabbro, D. Garlisi, I. Tinnirello, and M. Rossi, "A CSI dataset for wireless human sensing on 80 MHz Wi-Fi channels," *IEEE Communications Magazine*, vol. 61, no. 9, pp. 146–152, 2023.



Fig. 4. Class-wise fused recall on the fully held-out S1c recording. The time-shift control separates augmentation from the contrastive objective; both added methods retain poor jumping recall.

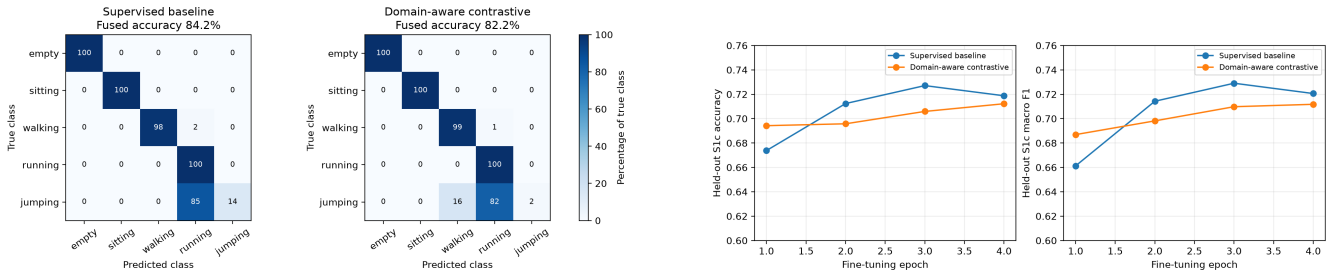


Fig. 5. Held-out S1c comparison. Left: fusion improves both methods, but the contrastive model retains a strong jumping–running confusion. Right: validation curves over fine-tuning epochs; the contrastive variant does not exceed the matched supervised baseline.